CAPSTONE PROJECT OVERVIEW

Fit-Life AI

Fit-Life AI is a health-and-fitness web platform designed to combine profile-driven wellness planning, AI-assisted food scanning, adaptive workouts, tracking, and recommendations into one practical daily-use product. The project aims to feel like a modern digital coach rather than a basic CRUD dashboard.

AI-first

Food analysis, recommendations, and planning experience

Profile-driven

Workouts and guidance adapt using BMI, goals, and health data

Mobile-ready

Responsive product flow for daily use on phones and desktop

Project Ideology

The core ideology of Fit-Life AI is to make health technology feel personalized, actionable, and motivating. Instead of showing isolated metrics, the platform connects user profile data, workouts, food habits, and recommendations into one guided experience.

The product is built around the idea that users should not need separate apps for calories, workouts, reminders, progress, and AI help. The system should work as an integrated assistant that understands goals, body condition, and daily behavior.

Core Product Principles

- Personalization over generic plans
- Usability over complex clinical presentation
- AI as guidance, not visual decoration
- Mobile-first experience for real daily usage
- Actionable tracking instead of passive dashboards

Present Frontend Tech Stack

HTML5 | CSS3 | Vanilla JavaScript | Responsive CSS Grid | Flexbox | Local Storage | REST API Integration | JWT-based Auth Flow | Render / Vercel Deployment

Frontend Architecture Style

The frontend is implemented as a multi-page web application using plain HTML, CSS, and JavaScript without a build system. Each major feature area is organized as a dedicated page and paired with a feature-specific script. Shared behavior such as API access, auth handling, sidebar navigation, toasts, and theme logic is centralized in reusable JavaScript modules.

Main Frontend Modules

- **Authentication:** welcome, login, register, and guarded page flow
- **Dashboard:** BMI, calories, water, streak, quote, and weekly insights
- **Health Profile:** onboarding-style profile capture and revisit summary
- **Food Scanner:** AI-style camera/upload flow with nutrition and warnings
- **Workout:** goal-based workout display, media blocks, timer, and progress flow
- **Tracker:** meals, workouts, water, sleep, and activity summary
- **Recommendations / AI Planner:** personalized guidance and chat-like support

Workflow of the System

1

User Onboarding and Auth

The user lands on the branded welcome screen, signs in, and is redirected into the protected app flow with session-aware navigation and token validation.

2

Health Profile Capture

The user enters personal health information such as age, gender, height, weight, activity level, and fitness goal. This becomes the decision base for BMI, calorie, body-fat, and workout logic.

3

Backend Analysis and Plan Generation

The backend uses stored profile signals to return personalized workouts, recommendations, dashboard summaries, and AI-assisted scanner output. Frontend pages then render these responses with safe fallbacks.

4

Daily Use Features

The user scans meals, reviews nutritional feedback, logs food intake, follows workouts, tracks progress, and receives recommendations and reminders through one connected interface.

5

Continuous Adaptation

When profile details change, workout and recommendation screens refresh to reflect the new goal path, making the product adaptive rather than static.

Major Frontend Highlights

- Branded Fit-Life AI landing experience with onboarding-style authentication flow
- Mobile-responsive UI across dashboard, scanner, workout, tracker, and recommendations pages
- Defensive API rendering so pages degrade gracefully when backend data is partial
- AI-style food scanner with preview, analysis states, warning banners, and meal logging
- Profile-driven workout page with hero section, activity cards, timers, and weekly plan layout
- Local persistence for supportive UX flows such as scan history, chat context, and cached profile state
- Clear separation between shared utilities and feature-specific page scripts

Why This Project Matters

Fit-Life AI demonstrates how a modern capstone project can connect health data, AI-assisted interaction, personalization, and responsive frontend engineering into one cohesive product. It is not only a website; it is a user journey centered on motivation, habit support, and informed daily wellness choices.

From a technical perspective, the project shows practical frontend architecture, API integration, state handling, responsive design, and real-world product thinking. From a product perspective, it shows how AI can support health workflows in a way that feels helpful and structured for end users.

Fit-Life AI • Project overview prepared for presentation and documentation use